The Infinite Loop Considered Harmless

The lambda calculus derived from itself, over one symbol

John Henry

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The bent card. The accomplice marks one card so the mark believes the quarter can be followed; this is that card, and the throw unbends it.

Reading note

The previous card said that the formal grammar of the construction has one terminal and that reading that terminal as a successor gives arithmetic. This card writes the calculus in which both facts are theorems, derives the calculus from itself in the standard way, exhibits the term that never stops, and proves that it is harmless in a precise sense: its record is the one-symbol tape, its reading is the counter, and it writes no quarter. Grades are as in the companion card: *proved here*, *standard*, *measured* (a count taken from a run of the middle card, not a theorem), and *proposed*. Section 4 runs the loop on purpose, as a search, and records what the middle card's own search leaves free. In the game this is the bent card: the one with a corner turned up so the mark thinks he can follow the quarter under it. The bend is a receipt supplied from outside, not a reading of the card, and the throw unbends it. Here the bend is the count of demonstrations, the five cards read as a stack; Proposition 3.3 is the throw.

1 The calculus from one symbol

Definition 1.1 (Terms)**.** Terms are generated by three rules: a variable x is a term; if M is a term and x a variable then $\lambda x.M$ is a term (abstraction); if M and N are terms then MN is a term (application). The one computation rule is β: $(\lambda x.M)N \to M[x := N]$ [3, 1]. *Standard.*

Definition 1.2 (Numerals)**.** For $n \geq 0$, $\overline{n} = \lambda f.\lambda x.\, f^n x$, where $f^n x$ is f applied n times to x. *Standard.*

Proposition 1.3 (The one-terminal grammar is the numerals)**.** *Let the grammar of the companion card have terminal* T *and let* T *be read as application of* f*. Then every sentence of the grammar whose nonterminals are all erased is* $f^n x$ *for the number* n *of terminals it contains, and the reader that counts terminals is the map* $\overline{n} \mapsto n$ *obtained by applying* $\overline{n}$ *to the successor and zero.* Proved here.

Proof. Erasing nonterminals leaves a string of n copies of T; reading each as one application of f gives $f^n x$. Applying $\overline{n}$ to succ and 0 gives $\mathrm{succ}^n 0 = n$. □

The abstraction λf is the grammar requiring a name for its symbol. The application that supplies succ is the choice. The numeral says nothing about what f is; that is handed in.

Proposition 1.4 (Two readings, two arithmetics)**.** *Addition of numerals is* $\lambda m.\lambda n.\lambda f.\lambda x.\, mf(nfx)$ *and multiplication is* $\lambda m.\lambda n.\lambda f.\, m(nf)$*. Reading* T *by tally uses only the first; reading a string of* n *terminals positionally in base two, as* $2^n - 1$*, uses the second.* Standard *for the definitions; the identification of the two readings with the companion card's tally and register is* proved here *by Proposition 1.3.*

2 The calculus from itself

A term can be written as a term, and a term can evaluate that term. This is the self-interpreter, and the construction below is Mogensen's [7]. *Standard; reproduced here.*

Definition 2.1 (Encoding)**.**

$$\ulcorner x \urcorner = \lambda a.\lambda b.\lambda c.\, a\, x, \qquad \ulcorner MN \urcorner = \lambda a.\lambda b.\lambda c.\, b \ulcorner M \urcorner \ulcorner N \urcorner, \qquad \ulcorner \lambda x.M \urcorner = \lambda a.\lambda b.\lambda c.\, c\, (\lambda x. \ulcorner M \urcorner).$$

Definition 2.2 (Fixed-point combinator)**.** $\mathbf{Y} = \lambda f.(\lambda x.f(xx))(\lambda x.f(xx))$, so that $\mathbf{Y}f \to f(\mathbf{Y}f)$ for every f.

Theorem 2.3 (Self-interpretation)**.** *Let*

$$\mathbf{E} = \mathbf{Y}\,(\lambda e.\lambda m.\, m\,(\lambda x.x)\,(\lambda p.\lambda q.\,(e\,p)(e\,q))\,(\lambda p.\lambda v.\, e\,(p\,v))).$$

Then $\mathbf{E} \ulcorner M \urcorner =_\beta M$ *for every term* M*.* Standard *[7].*

Proof. Induction on M. For a variable, $\mathbf{E}\ulcorner x \urcorner \to \ulcorner x \urcorner(\lambda x.x)(\dots)(\dots) \to (\lambda x.x)x \to x$. For an application, $\mathbf{E}\ulcorner MN \urcorner \to (\mathbf{E}\ulcorner M \urcorner)(\mathbf{E}\ulcorner N \urcorner) =_\beta MN$ by the hypothesis. For an abstraction, $\mathbf{E}\ulcorner \lambda x.M \urcorner \to \lambda v.\mathbf{E}(\ulcorner M \urcorner[x := v])$, and $\ulcorner M \urcorner[x := v] = \ulcorner M[x := v] \urcorner$ since the encoding of a variable is the variable itself under a; the hypothesis gives $\lambda v.M[x := v]$, which is $\lambda x.M$ up to renaming. □

This is what "the calculus derived from itself" means: the evaluator is a term of the calculus, the syntax is a term of the calculus, and the theorem that one applied to the other returns the original is a theorem of the calculus. Nothing outside it was used. The one thing it does not do is choose f: the self-interpreter evaluates numerals faithfully and leaves their abstraction abstract.

3 The loop

Definition 3.1 (The term that never stops)**.** $\mathbf{\Omega} = (\lambda x.xx)(\lambda x.xx)$. Then $\mathbf{\Omega} \to \mathbf{\Omega}$, and $\mathbf{\Omega}$ has no normal form [1]. *Standard.*

The self-interpreter inherits it: $\mathbf{E}\ulcorner \mathbf{\Omega} \urcorner$ reduces forever, and $\mathbf{Y}$ itself is built from the same self-application. Dijkstra's objection to the unrestricted jump [4] was that it severs the record of a computation from the text that describes it: after a jump, where the process is cannot be read from where the program is. The loop is the jump that jumps to itself. The title claims it is harmless, and the claim has three parts.

Proposition 3.2 (The loop's record is the one-symbol tape)**.** *Read each reduction step as writing one terminal. Then the record of* $\mathbf{\Omega}$ *after* n *steps is the string* T^n*, the tally reads* n*, and the positional reading is* $2^n - 1$*. No step writes a different symbol, and no prefix of the record determines any later step differently from the first.* Proved here.

Proof. Each step is the same β-redex reducing to itself; by Proposition 1.3 the step record is one terminal per step. The last clause is the statement that $\mathbf{\Omega} \to \mathbf{\Omega}$ is the only step. □

So the loop is the counter and nothing else: a record that grows by one unchanged symbol per step, whose positional reading grows exponentially while the thing read never changes. That is the de Sitter counter of the numerical card, exactly, and it is harmless for the same reason: it changes nothing but the count.

Proposition 3.3 (The loop writes no quarter)**.** *No prefix of the record of* $\mathbf{\Omega}$ *determines a value of any free variable, and* $\mathbf{\Omega}$ *has none. In particular the loop supplies no* f *to any numeral and decides no comparison.* Proved here.

Proof. $\mathbf{\Omega}$ is closed, and its reduction sequence contains only $\mathbf{\Omega}$; a record consisting of one closed term repeated determines nothing about any variable. □

Proposition 3.4 (Harm is a loop read as a value)**.** *Let a record system accept a term as the value of a computation only when the computation has reached a normal form. Then* $\mathbf{\Omega}$ *is never accepted as a value, and a system that accepted it would accept every term, since for every* M *the term* $(\lambda x.M)\mathbf{\Omega}$ *reduces to* M *along one path and loops along another.* Proved here.

Proof. The first clause is the definition. For the second, if a looping reduction counted as reaching a value, then the looping path of $(\lambda x.M)\mathbf{\Omega}$ would certify a value for a term whose other path certifies M, for arbitrary M: every term would be a value of every other. □

That is the only harm the loop can do, and it is done by the reader, not by the loop. The loop considered as a record is a counter. The loop considered as a value is a contradiction. A record system stays consistent by keeping the arm that says "not yet" separate from the arm that says "done": the computation that has not halted is absence of evidence, and reading it as evidence of absence is the one mistake. The typed calculus in which the middle card is written makes this structural: self-application is not well typed there, $\mathbf{Y}$ has no type, every term normalizes, and the loop cannot be written at all. The price is that fixed points must be handed in from outside, which is the quarter in its λ-calculus form: $\mathbf{Y}f = f(\mathbf{Y}f)$ supplies a fixed point by a combinator in the untyped calculus and by a choice in the typed one.

4 The search outside the record

The loop is a computation that never halts and is harmless because it is never read as a value. A search is the same thing run on purpose: it may not halt, and it is harmless for the same reason, as long as only what it finds is read.

Proposition 4.1 (A complete search confirms and does not exhaust)**.** *A sentence of first-order logic is unsatisfiable if and only if some finite set of its ground instances, terms built from its own symbols, is already unsatisfiable. So searching the ground instances in order halts whenever there is a refutation to find, and in general runs forever when there is none.* Standard *[5].*

The space of ground instances is the whole space the sentence can say anything about. A search through all of it is the loop with a stopping condition: it writes one instance per step, and it stops only when what it has written closes.

Proposition 4.2 (Search and receipt are separate)**.** *A search that halts can hand a checker a certificate, and the checker reads the certificate without repeating the search. The value of the fifth Busy Beaver function was settled this way: programs enumerated the machines and decided each one, and a proof assistant checked the certificates [9].* Standard. *The middle card is built the same way: an elaborator searches, writes what it found as literals, and the checker confirms each line against the one before.* Measured*, from the card's construction.*

So the search is never in the space the checker reads. Its results are; its process is not. That is the loop kept apart from the value, as in Proposition 3.4, now for a loop that was meant to stop.

Forced and free. The middle card's steps carry universe levels, one per seat. Raising each seat by itself and checking whether the step still holds sorts the seats in two. Most are forced: the checker accepts one level only. A few are free: any level is accepted, and nothing in the step reads them. In the thirteenth step through the thirtieth, exactly two seats are free in each. From the step that first receives the local model, free seats accumulate and are inherited: four, five, seven, eight, nine, nineteen, thirty-five, nineteen. Raising every free seat of a step at once, each to a different level, is accepted too, so the free seats are free together. One hundred and forty-nine are free in all. *Measured*, from a run of four thousand six hundred and thirty-five single-seat changes. The forced seats are themselves computed from the description: each forced lift is the universe one construction keeps its quantum in, set once and carried, and reading the steps for it reproduces the hand census on every step it covers. *Measured.*

So each step's space is one forced point times every assignment of its free seats. The search the device runs visits only the forced point. The complete space, the free seats ranging over every level, is what a complete search would have to cover, and by Proposition 4.1 it could confirm a member and never exhaust them. The shortest member, every free seat at the lowest level, is the one the device writes down: a Berry description of a point in a space no search finishes.

The flat region. The sinc integral $\int_0^\infty \prod_{k=0}^{n} \operatorname{sinc}(x/(2k+1))\,dx$ equals $\pi/2$ while $1/3 + 1/5 + \cdots + 1/(2n+1) < 1$, which holds through the factor $1/13$ (0.955) and fails at $1/15$ (1.022) [2]. *Standard; the sums are proved here.* The proof runs through convolution: each sinc transforms to a box, the product to a convolution of boxes, and the integral is that convolution's value at the origin, flat while the narrower widths fit inside the first. Repeated convolution of boxes builds splines; integration by parts, which is Stokes' theorem on a line, moves a derivative onto the test function and gives the weak form that spline elements solve; the natural cubic spline through given points is the curve of least bending energy [6], whose Euler–Lagrange equation is the beam equation, with fourth derivative zero where unloaded. *Standard*, each step. Four boxes convolved are the cubic spline, and four is the count at which the companion cards read once around.

5 Three bent cards

The reading note calls the bent card a receipt supplied from outside, and the paragraph after Proposition 3.4 says why a typed calculus needs one: its fixed points cannot be computed by self-application and must be handed in. The middle card, run to its end, needs exactly three, and each is the step its loop cannot take for itself.

The base case of the search. A carrier holds a slot it never reads. Fold a tape onto the slot and the record is the tape followed by the slot, so a reader that subtracts the slot's length reads the tape's length, whatever the slot holds. *Proved here*: the length of a concatenation is the sum of the lengths. Handed that receipt, a walk read through a carrier of carriers is the walk already stored for the inner card, and the check is a lookup, finished in under a second. *Measured.* Without it, a reader that keeps the slot reads every length one cell long. On the walk's quadratic step this changes the coefficients from $(13, 18, 2)$ to $(14, 19, 3)$, and the third no longer makes the step divide by its derivative. The fixed points become $(57 \pm \sqrt{1121})/56$, the upper one repels with multiplier near -2.08, the halting test never holds, and the integers the step keeps double their length every turn. *Proved here* for the algebra. *Measured*: two runs of that reader were stopped by hand, one at

92 GB and one at 18 GB, with no reading. The loop is harmless exactly when the receipt is handed in, as Proposition 3.4 says of the loop read as a value.

The halting test. The walk stops when its correction falls below the last printed place, and from there it holds the same pair. By the exact error law of the derivative's step, the remaining error is $e^2/(2e + d)$, which is not zero; the test declares it zero. *Proved here*, from the error law. The step the test stands for, continuing to the limit, would land on a root containing $\sqrt{10}$, which no finite record holds. Written down exactly, it is a real number, and the reals of the middle card's language are built as classes of sequences whose operations are not computable: naming the root exactly costs the choice the card's reading never spends. *Standard* for the construction of the reals; that the halting test is the declared fixed point is *proposed*. A difference declared a sameness: the true-equals-false of the argument, and the second bent card.

The calibration. The reading is a ratio of counts. To give it units, let one symbol be one tick of the caesium hyperfine line; then the speed of light is one symbol per tick. Its digits in metres per second were declared exact in 1983, and the Planck constant and the elementary charge in 2019 [8]. *Standard*. With those three declarations handed in, the reading fixes the one electromagnetic constant still measured: the impedance of free space, which the reading puts at $376.798\,\Omega$ against a measured $376.730\,\Omega$, while the speed of light, the product of the two vacuum constants, does not move. *Proved here* as arithmetic. Three finite strings declared exact are the third bent card.

Proposition 5.1 (What no bent card absorbs)**.** *With all three receipts handed in, the reading is the same under every reader that measures length, every slot, and every assignment of the free seats; under the declared constants it appears entirely in the impedance of free space; and there it differs from measurement by* 1.80 *parts in ten thousand, which measurement excludes at about one part in* 10^{10}.

Proved here for the first clause from the subtraction above and the invariances of the companion cards; *measured* for the free seats, every free seat of steps fourteen through thirty-eight having been moved off its value at once with the steps still checking alike; *proved here* for the impedance; *standard* for the exclusion.

So the argument closes on the bent card, and closes this way. The middle card is a finite, checked description, and every step it cannot take for itself is a receipt it names: the base case of its search, the zero its halting test declares, and the constants that give it units. Each is a fixed point handed in from outside. The residue is the one thing none of them absorbs, and it is the card's strongest line: no convention, no declaration of exactness, and no choice of reader moves it. That the residue is physical, and not the description's own, is the knower's.

6 What is proposed

Remark (The counter is the loop)*.* The companion cards carry a counter that increases by exactly one per written entry, a register that is the positional reading of a one-symbol tape, and a verdict decided once at the top. Propositions 3.2 and 3.3 identify the counter with the record of $\mathbf{\Omega}$ read one terminal per step: the publication's growth is the loop's growth, harmless because the loop's symbol never changes. The further identification of the loop with the global time of the de Sitter slicing, and of its exponential positional reading with the scale factor, is *proposed*, as it was in the

numerical card. What would discharge it is unchanged: the cosmological constant as a residue is derived; the map from the record's steps to a physical clock is not.

Remark (The free seats). That the free seats are the weakly interacting part of the construction: present in every instance, read by none, and growing once the local model is handed in. That the first thirteen steps are the flat region of the sinc integral, exact up to the factor 1/13, so that a search may take them as computed and begin at the fourteenth. That the device's least action, closing at degree three, is the bending energy of an elastic strip through its knots. *Proposed*, each: the counts agree, and no map between the two sides is derived.

Remark (Considered harmless). Dijkstra was right about the jump, and the loop is the jump that is harmless, for one reason: it never leaves the text. Where the process is and where the program is coincide at every step, because the step is the program. A loop that wrote a second symbol would be a computation; a loop that wrote the first symbol forever is a clock; and a clock is the one thing a record cannot be wrong about, since it asserts nothing but that another entry was written. The harm begins at the first reader who takes the clock for a value. The middle card's press check reads once, at the top, and decides exclusive-or between two readings; it never reads the loop, because the loop has no top.

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